



GUIDELINES FOR GEFORCE RTX TECHNOLOGIES

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NVIDIA DLSS

[NVIDIA DLSS](#) is a neural graphics technology that multiplies performance using AI to create entirely new frames, display higher resolution through image reconstruction, and improve the image quality of intensive ray-traced content—all while delivering best-in-class image quality and responsiveness.

DLSS 4.5 is a suite of technologies comprised of DLSS Dynamic Multi Frame Generation, DLSS Frame Generation, DLSS Super Resolution, DLSS Ray Reconstruction, NVIDIA DLAA, NVIDIA Reflex Low Latency and delivers boosted frame rates, great responsiveness, and better than native IQ. To deliver this intended gaming experience, it is important for users to be able to easily toggle all components of DLSS with one global switch.

- **DLSS Multi Frame Generation** uses AI to boost frame rates by generating up to 5 additional high-quality frames, all while optimizing responsiveness with [NVIDIA Reflex](#). Also includes Dynamic Multi Frame Generation, which dynamically adjusts frame generation multiplier to achieve target frame rate. DLSS Multi Frame Generation is available for GeForce RTX 50 Series GPUs.
- **DLSS Frame Generation** uses AI to boost frame rates by generating one additional high-quality frame, all while optimizing responsiveness with [NVIDIA Reflex](#). DLSS Frame Generation requires a GeForce RTX 40 Series GPU or higher and a Reflex SDK integration.
- **DLSS Super Resolution** boosts frame rates by rendering fewer pixels and using AI to reconstruct high resolution frames. This feature is available for all RTX GPUs.
- **DLSS Ray Reconstruction** uses AI to enhance image quality and generate additional pixels for intensive ray-traced scenes. Ray Reconstruction replaces hand-tuned denoisers with an NVIDIA supercomputer-trained AI network that generates higher-quality pixels in between sampled rays. This feature is available for all RTX GPUs and requires Super Resolution to be enabled.
- **NVIDIA DLAA** maximizes image quality through an AI-based anti-aliasing technique. DLAA uses the same Super Resolution technology developed for DLSS, reconstructing a native resolution image to maximize image quality. This feature is available for all RTX GPUs.
- **NVIDIA Reflex** helps DLSS Multi Frame Generation and Frame Generation achieve maximum performance at the lowest latency. It accomplishes this by synchronizing the CPU frame submission with GPU processing for just in time rendering.

UI Tooltip or Setting Description

- **NVIDIA DLSS:** NVIDIA DLSS uses AI to boost frame rates while maintaining great image quality and responsiveness. This feature requires a GeForce RTX graphics card.
 - **Frame Generation:** Frame Generation boosts frame rates by using AI to render additional frames. 2X is supported on a GeForce RTX 40 Series graphics card or higher, while Fixed and Dynamic through 6x are supported on a GeForce RTX 50 Series graphics card.
 - **Super Resolution:** Super Resolution boosts frame rates by rendering fewer pixels and using AI to output high resolution frames. This feature requires a GeForce RTX graphics card.
 - **Ray Reconstruction:** Ray Reconstruction enhances image quality by generating high-quality pixels between sampled rays for intensive ray traced content. This feature requires a GeForce RTX graphics card.
 - **DLAA:** DLAA is an AI-based anti-aliasing mode to improve image quality. This feature requires a GeForce RTX graphics card.

- **NVIDIA Reflex Low Latency** - *NVIDIA Reflex reduces system latency and increases PC responsiveness.*

UI Menu Options

When DLSS Frame Generation is ON, please enable NVIDIA Reflex Low Latency.

- DLSS Frame Generation 2x is available for RTX 40 Series RTX graphics cards or higher.
- DLSS Frame Generation Dynamic, 3x - 6x are available for RTX 50 Series RTX graphics cards.

When DLSS Super Resolution is ON, please disable scaling technologies (Resolution Scaling, NVIDIA Image Scaling), and Anti-Aliasing. DLSS Super Resolution is available for all RTX graphics cards.

UI: DLSS ON

NVIDIA DLSS	◀ ON ▶	OFF, ON (Default)
Frame Generation	◀ Dynamic ▶	OFF, Dynamic (Default), 2X, 3X, 4X, 5X, 6X
Target Frame Rate	◀ 240 FPS ▶	Range between [100 FPS - Highest Refresh Rate (Default)]
Super Resolution	◀ AUTO ▶	OFF, Auto (Default), Quality, Balanced, Performance, Ultra Performance, Custom, DLAA
Ray Reconstruction	◀ ON ▶	ON, OFF
Sharpness	0 	0 (Default), 0 TO 1 (Value dependent on developer choice)
NVIDIA Reflex Low Latency	◀ ON ▶	ON (Default), Boost, OFF (OFF not available if Frame Generation is enabled)
Frame Warp	◀ OFF ▶	OFF (Default), ON (Also enables NVIDIA Reflex Low Latency)
NVIDIA Image Scaling	◀ DISABLED ▶	Feature is disabled when DLSS Super Resolution is ON
Sharpness	0 	Feature is disabled when DLSS Super Resolution is ON
Anti-Aliasing	◀ DISABLED ▶	Feature is disabled DLSS Super Resolution is ON
Resolution Scale	100 	Feature is disabled DLSS Super Resolution is ON
VSYNC	◀ DISABLED ▶	Feature is disabled DLSS Frame Generation is ON

Developer UI Checklist

- **DLSS Feature Support**
 - We **strongly** recommend a global DLSS Feature toggle, which enables DLSS Frame Generation, DLSS Super Resolution, Ray Reconstruction, and NVIDIA Reflex Low Latency mode. This is the easiest way for end users to get the targeted performance, image quality, and latency benefits of DLSS.
 - Please make sure NVIDIA DLSS is **disabled** when on unsupported hardware or drivers.
 - Please make sure PCL Markers are always enabled on all vendor GPUs, whether Reflex is toggled on or off. This allows all gamers to accurately measure their PC latency, even on non NVIDIA GPUs.
- **DLSS Frame Generation**
 - DLSS "Auto" is available for RTX 40 Series RTX graphics cards and set to 2x by default.
 - DLSS "Dynamic" is available for RTX 50 Series RTX graphics cards and set to 6x by default.
 - DLSS Frame Generation **requires** NVIDIA Reflex Low Latency integration and enablement. Users should not be able to disable NVIDIA Reflex Low Latency when Frame Generation is ON.

- DLSS Frame Generation **requires** Windows OS “Hardware-accelerated GPU Scheduling” and fails to initialize if not enabled. If this occurs, the game should notify the user to enable this feature.
- **DLSS Super Resolution**
 - The DLSS mode titled “Auto” should be the first option in the UI after Off and be enabled by default when NVIDIA RTX hardware is detected. The Auto mode isn’t itself a true mode and should select the appropriate default mode from the table below depending on the current output resolution.
 - When the UI shows the DLSS modes horizontally or in a left-right scrolling list, the order should be: Off, Auto, Quality, Balanced, Performance, Ultra-Performance, DLAA.
 - When the UI shows the DLSS modes vertically or in an up-down scrolling list, the order should be:
 1. Off
 2. Auto
 3. Quality
 4. Balanced
 5. Performance
 6. Ultra-Performance
 7. DLAA
 - When DLSS Super Resolution is turned on, **make sure the anti-aliasing settings are disabled** (both UI, as itself)
 - When DLSS Super Resolution is turned on, **make sure the resolution scale settings are disabled** (UI disables, the application uses render target size from DLSS optimal settings)
- **DLSS Ray Reconstruction**
 - The DLSS mode titled “Auto” should be the first option in the UI after Off, and be enabled by default when NVIDIA RTX hardware is detected, DLSS Super Resolution is enabled, and when Ray Tracing content is enabled.
 - **Requires DLSS Super Resolution to be ON.** Disable when Super Resolution is not enabled.
 - Requires Ray Tracing to be ON. Disable when Ray Tracing is not enabled.
- **Sharpness Slider**
 - Sharpness slider is a requirement for DLSS integration and the preferred integration is NVIDIA Image Scaling (please see below). When sharpness is on for DLSS, make sure other sharpening features are disabled.

DLSS Multi Frame and DLSS Frame Generation Options

Mode	Description	Resolution Support	GPU Requirements
Off	Turn DLSS Frame Generation off.	N/A	N/A
Auto	Enables Frame Generation as needed to 2X to boost performance	ALL RESOLUTIONS	RTX 40 Series RTX GPUs ONLY
Dynamic	Dynamically enables Multi Frame Generation modes to target FPS.	ALL RESOLUTIONS	RTX 50 Series RTX GPUs

Target Frame Rate (OPTIONAL)	Optional control for Dynamic for 120Hz+ monitors. Sets target frame rate between 100 FPS to highest refresh rate (Default value) at current resolution. ie. [100 FPS - 480 FPS]	ALL RESOLUTIONS	RTX 50 Series RTX GPUs
2X	The total number of frames is doubled, with one rendered and one generated frame.	ALL RESOLUTIONS	RTX 40+ Series RTX GPUs
3X	Generates two additional frames for one rendered for a total of three frames.	ALL RESOLUTIONS	RTX 50 Series RTX GPUs
4X	Generates three additional frames for one rendered for a total of four frames.	ALL RESOLUTIONS	RTX 50 Series RTX GPUs
5X	Generates four additional frames for one rendered for a total of five frames.	ALL RESOLUTIONS	RTX 50 Series RTX GPUs
6X	Generates five additional frames for one rendered for a total of six frames.	ALL RESOLUTIONS	RTX 50 Series RTX GPUs

DLSS Super Resolution Options

Mode	Description	Resolution Support	GPU Support
OFF	Turns DLSS Super Resolution off.	N/A	N/A
AUTO	Selects the best DLSS Mode for the current output resolution.	ALL RESOLUTIONS	ALL RTX GPUs
QUALITY	Offers higher image quality than balanced mode.	ALL RESOLUTIONS	ALL RTX GPUs
BALANCED	Offers both optimized performance and image quality.	ALL RESOLUTIONS	ALL RTX GPUs
PERFORMANCE	Offers a higher performance boost than balanced mode.	ALL RESOLUTIONS	ALL RTX GPUs
ULTRA PERFORMANCE	Offers the highest performance boost.	ALL RESOLUTIONS	ALL RTX GPUs
DLAA	Recommended placement for DLAA.	ALL RESOLUTIONS	ALL RTX GPUs

These are the recommended DLSS settings based on output resolution. This is automatically set when user sets to 'AUTO'

Default DLSS Mode	Output (Resolution)	Output (Megapixels)
Disabled	Below 1920x1080	Below 2.03
Quality mode	Equal to 1920x1080, equal or below 2560x1440	Up to 3.68
Performance mode	Greater than 2560x1440, equal or below 3840x2160	3.69 - 8.29
Ultra Performance mode	3840x2160 and higher (e.g. 5120x2880 and 7680x4320)	3.69+

DLSS Ray Reconstruction Options

Mode	Description	Resolution Support	GPU Support
OFF	Turns Ray Reconstruction OFF	N/A	N/A
AUTO	Automatically enables Ray Reconstruction when Ray Tracing is ON, or disables it when Ray Tracing is OFF	ALL RESOLUTIONS	ALL RTX GPUs

DLSS Super Resolution & Dynamic Resolution Systems

As detailed in the DLSS Programming Guide, DLSS Super Resolution can support dynamically varying input sizes if the renderer has a Dynamic Resolution System (DRS).

If DRS is enabled, the game UI should:

1. Present only two options for DLSS Super Resolution: "Off" and "On"; or
2. If the UI system does not allow the DLSS Super Resolution options to change, disable (hide or gray out) all DLSS Modes and only allow the user to select "Off" or "Auto".

NVIDIA REFLEX

The Reflex SDK is an easy-to-integrate set of NVAPIs that provide both system latency optimization and measurement capabilities, giving players a more responsive feel in-game. The SDK strives to keep players at the lowest possible latency by dynamically adjusting submission timing of rendering work to the GPU so that they are processed just-in-time. The SDK includes the ability to measure PC Latency and automatically configure Reflex Analyzer which helps gamers optimize settings. Reflex also helps DLSS Frame Generation achieve maximum performance in addition to reducing latency.

UI Tooltip or Setting Description

- **NVIDIA Reflex Low Latency** - *NVIDIA Reflex reduces system latency and increases PC responsiveness.*

UI Menu Options

Low Latency Mode - "NVIDIA Reflex Low Latency"

General Display **Graphics** Stats

Frame Rate Limiter < Unlimited >

NVIDIA Reflex Low Latency < On >

Overall Graphics Quality < Medium >

Options are Off,
On, On + Boost

Note: Reflex is required for DLSS Frame Generation, and will automatically turn on when DLSS Frame Generation is enabled. Reflex does not currently support other frame generation technologies (e.g. FSR), and should therefore be unavailable when these are enabled.

Settings Options

NVIDIA Reflex Low Latency

Mode	Description	Driver Support	GPU Support
Off	Low Latency mode is disabled.	N/A	N/A
On	Low Latency mode is now enabled and optimizes system latency.	R455+	Maxwell GPUs+ (Pascal+ for VK)
On+ Boost	Low Latency mode is now enabled and optimizes system latency. In Boost mode, NVIDIA Reflex will save slightly more latency and attempt to optimize latency in CPU bound cases. This option can slightly decrease FPS and increase GPU power draw.	R455+	Maxwell GPUs+ (Pascal+ for VK)

Settings Defaults

PC Latency is supported for all IHVs and should be **enabled at all times**. No UI setting is required

NVIDIA IMAGE SCALING

NVIDIA Image Scaling enables developers to complement their NVIDIA DLSS integrations, so they can provide the best image quality with NVIDIA DLSS, and cross-platform support with NVIDIA Image Scaling. NVIDIA Image Scaling offers best-in-class spatial scaling and sharpening and supports all GPUs.

Naming Guidelines

There is no strict requirement to brand this as an NVIDIA feature. Here are optional recommendations for UI names:

- “NVIDIA Image Scaling”
- “Image Scaling”

UI Tooltip or Setting Description

- *NVIDIA Image Scaling boosts frame rates using GPU scaling and sharpening.*

UI Menu Options

Developers can expose NVIDIA Image Scaling either as a standalone UI element or combined with other performance boosting features (NVIDIA DLSS, etc)

Standalone UI: NVIDIA Image Scaling

NVIDIA IMAGE SCALING<AUTO>

SHARPNESS0

RESOLUTION SCALE100

AUTO (Default), ULTRA QUALITY, QUALITY, BALANCED, PERFORMANCE, CUSTOM, OFF
0 (Default, Value dependent on developer choice), 0 TO 1
Feature is disabled when NVIDIA Image Scaling is ON

Combo UI: NVIDIA Image Scaling

PERFORMANCE BOOST<NVIDIA IMAGE SCALING>

IMAGE SCALING MODE<AUTO>

SHARPNESS0

RESOLUTION SCALE100

AUTO (Default), ULTRA QUALITY, QUALITY, BALANCED, PERFORMANCE, CUSTOM, OFF
0 (Default, Value dependent on developer choice), 0 TO 1
Feature is disabled when NVIDIA Image Scaling is ON

Mode Defaults

These are the default NVIDIA Image Scaling Mode settings based on output resolution:

Quality Preset	Scale Factor	Input Resolution for 2160P Output	Input Resolution for 1440P Output	Input Resolution for 1080P Output
Ultra Quality	77%	2954x1662	1970x1108	1477x831
Quality	66.667%	2560x1440	1706x960	1280x720

Balanced	59%	2259x1271	1506x847	1129x635
Performance	50%	1920x1080	1280x720	960x540
Custom	50%-100%	1920x1080 to 3840x2160	1280x720 to 2560x1440	960x540 to 1920x1080

RAY TRACING

Ray tracing is the holy grail of gaming graphics, simulating the physical behavior of light. [GeForce RTX graphics cards](#) have dedicated RT Cores to accelerate ray tracing, enabling higher quality and performance.

Additional details on Ray Tracing Best Practices can be found [here](#).

Ray Tracing: Settings Options

The recommended RT settings are “ON”, “HIGH”, and “ULTRA.” When ray tracing is “ON,” there should be a noticeable image quality difference. Additionally, there should be a very noticeable difference between each quality level, otherwise fewer setting options are appropriate. In the case of path tracing, individual Ray Tracing effects settings such as Shadows or Reflections should be grayed out and a global Ray Tracing setting should be set to “ULTRA”.

Ray Tracing: Target Performance

We recommend the following targets for 60 fps average gameplay in your benchmark or areas of the game that are relatively heavy for ray tracing effects.

- **GeForce RTX 5060/4060 Ti:** Ray tracing set to “ON” at 1920x1080 with DLSS enabled
- **GeForce RTX 5070/4080:** Ray tracing set to “HIGH” at 2560x1440 with DLSS enabled
- **GeForce RTX 5080:** Ray tracing set to “ULTRA” at 2560x1440 or 4K with DLSS enabled

Ray Tracing: Recommended UI

DISPLAY **GRAPHICS** ADVANCED

DX12 ENABLED

OFF

ON

RAY TRACED [REFLECTIONS]

<

ON

>

RAY TRACED [SHADOWS]

<

ON

>

DXR requires DX12 to be enabled

OFF, ON, HIGH, ULTRA

OFF, ON, HIGH, ULTRA

UI Tooltip or Setting Description

- **DXR:** Enable DirectX Raytracing (DXR) for life-like [EFFECT NAME] (i.e., Shadows, Reflections, etc)
- **NON-DXR:** Enable ray tracing for life-like [EFFECT NAME] (i.e., Shadows, Reflections, etc)

RTX Hair

RTX Hair uses Linear-Swept Spheres (LSS), a GeForce RTX 50 Series GPU-accelerated algorithm for path-traced strand-based hair. RTX Hair reduces the amount of geometry necessary to render strands of hair, and uses spheres/tubes instead of triangles to get a more accurate fit for hair shapes in ray tracing.

Additional details on LSS can be found [here](#).

RTX Hair: Settings Options

The recommended RTX Hair settings are “ON” and “OFF”. When RTX Hair “ON”, there should be a noticeable lighting difference on individual strands of hair. RTX Hair requires all RT settings to be enabled.

RTX Hair: Recommended UI



UI Copy

- **RTX Hair:** More detailed hair with realistic lighting and shadows on GeForce RTX 50 Series GPUs.

NVIDIA ADAPTIVE SHADING (NAS)

NVIDIA Adaptive Shading (NAS) boosts performance by selectively lowering pixel shading rate, without affecting perceived image quality. Screen regions without high contrast details or with fast motion speeds are identified and shaded in lower rate, using the Variable Rate Shading (VRS) feature introduced on Turing.

Game Options

The recommended NAS settings are “OFF”, “BALANCED”, “PERFORMANCE, ” and CUSTOM.

UI Options

DISPLAY GRAPHICS **ADVANCED**

NVIDIA ADAPTIVE SHADING (NAS)

< BALANCED >

DETAIL SENSITIVITY

50%

LOW-LIGHT SENSITIVITY

50%

MOTION SENSITIVITY

50%

OFF, PERFORMANCE, BALANCED, QUALITY, AND CUSTOM

Notes:

NAS changes to Custom mode when user changes the default settings (Detail sensitivity, low-light sensitivity, and motion sensitivity).

UI Copy

- **NVIDIA ADAPTIVE SHADING:** *Boost frame rates by adapting shading rate based on content and motion information. This will disable deferred rendering.*
- **DETAIL SENSITIVITY:** *Shading rate sensitivity to image details*
- **LOW-LIGHT SENSITIVITY:** *Shading rate sensitivity to dark regions*
- **MOTION SENSITIVITY:** *Shading rate sensitivity to motion*

NVIDIA DYNAMIC VIBRANCE

RTX Dynamic Vibrance SDK is an AI network that transforms color to improve visual clarity, allowing players to better distinguish in-game elements. Dynamic Vibrance adaptively tunes color saturation and contrast depending on the game scene—colors pop more on-screen, and color crushing is minimized. Featuring adjustable intensity and saturation, Dynamic Vibrance provides players with a flexible option to calibrate the game’s visual aesthetic to their preference.

UI Tooltip or Setting Description

- RTX Dynamic Vibrance** - *RTX Dynamic Vibrance enhances visual clarity by dynamically tuning color saturation and contrast.*

UI Menu Options

General

Display

Graphics

Stats

RTX Dynamic Vibrance

< ON >

Intensity

0%

50%

100%

Saturation

0%

50%

100%

Note: Lowering “Intensity” to 0% is equivalent to turning Dynamic Vibrance off, and should therefore gray out the Saturation slider to make it inoperable.

Settings Options

Mode	Description	Driver Support	GPU Support
OFF	Turns Dynamic Vibrance off	N/A	N/A
ON	Turns Dynamic Vibrance on	r550+	Turing+ (20-series)
INTENSITY	Dictates the strength of the effect, where 100% is strongest, and 0% is equivalent to turning Dynamic Vibrance off.	r550+	Turing+ (20-series)
SATURATION	Dictates the range of colors affected, where 100% is strongest, and 0% is equivalent to no saturation boost.	r550+	Turing+ (20-series)

Settings Defaults

Dynamic Vibrance “Off” is the recommended default setting. When Dynamic Vibrance is enabled, the default recommended values for both intensity and saturation are 50%.

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